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Coping with climate shocks

A review of operationalizations of (im)mobility capabilities

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Abstract

1. Introduction

In migration studies, climate change gained relevance in the 1990s after an IPCC report suggested that one of the most significant consequences of the phenomenon could be its impact on human mobility (International Organization for Migration, 2008). Early studies projected up to 200 million internal “climate refugees” by 2050, while the updated Groundswell report places the figure at 216 million people (Clement et al., 2021; International Organization for Migration, 2008). Despite the earlier estimates, in 2011 the UK Government’s Foresight (2011) report, was the first influential document to argue against this deterministic view, and recent evidence supports this notion. For example, by the end of 2024, the IDMC (2025) estimated 83.4 million internally displaced people worldwide, of whom 73.5 million were displaced by conflict and violence, compared to 9.8 million by natural disasters. Furthermore, international migration rates have remained relatively stable since 1990, fluctuating around 1.13% to 1.29% of the global population every five years (Azose & Raftery, 2019). Jointly, these figures beg to question the assumption that climate change will mechanically lead to more mobility.

In line with this paradox, subsequent research has revealed a more complex relationship between climate change and mobility, indicating that migration (whether voluntary or not) is only one among several possible responses to climate change. In regard to this, Mallick & Hunter (2023) suggest that both migration and non-migration may function as households’ adaption strategies. This complexity is further reflected in studies highlighting that climate change may simultaneously increase the desire to migrate while reducing people’s ability to do so, thereby generating forms of involuntary immobility or “trapped” populations (Freihardt, 2025; Letta et al., 2024). Similarly, a systematic review focusing on the world’s drylands finds that climate-related events are associated with both mobility and involuntary immobility outcomes across different contexts in 76% of these studies (Link et al., 2025).

Taking the previous into account, an important challenge within migration research relates to how to understand different forms of (im)mobility. In this regard, one of the most prominent theoretical frameworks which has explicitly sought to explain these varied responses is de Haas’s (2021) aspirations-capability framework. According to this framework, migration is the result of both people’s aspirations to migrate and their capability to do so. Within this perspective, immobility could result either from limited aspirations to migrate, from constraints affecting people’s capability to move, or both. Focusing on the capability to migrate, the Foresight (2011)

report noted that impoverished people face a double risk by being both more vulnerable to environmental degradation but also unable to escape from it due to limited resources.

Contrary to the previous perspective, Schewel (2020) and de Haas, (2021) warn against assuming that all immobile people lack agency or mobility capabilities. Indeed, previous research has shown that even in environmentally exposed areas many households remain voluntarily due to factors such as place attachment (Adams, 2016). Similarly, Ahsan et al.'s (2021) study found that in Coastal Bangladesh 88% of respondents recognized themselves as voluntary non-migrants despite living in an environmentally exposed area. Furthermore, the authors also found significant differences between the voluntary and non-voluntary non-migrants. Compared to the later, voluntary non-migrants had higher socioeconomic and sociopsychological advantages as well as greater access to governmental and non-governmental support networks (Ahsan et al., 2021). These differences suggest that correctly differentiating between immobile populations that are effectively constrained from migrating and those that are not, is crucial, as the first may require, not only more, but also different forms of support and policy approaches than the later¹.

Turning back to the aspirations-capability framework, this creates an important methodological challenge regarding how to empirically identify people's aspirations and capabilities to migrate. While migration aspirations are often measured through individuals' self-reported desire to move (Freihardt, 2025; DeWaard et al., 2022), migration capabilities are typically treated as a latent construct inferred indirectly through proxy indicators. As a result, authors have operationalized mobility capabilities differently, varying from one-dimensional household or individual indicators (Adams, 2016; DeWaard et al., 2022) to multidimensional country-level (Rodríguez-Peña, 2025) or individual-level approaches (Hosnedlová & Maidanik, 2025; Obi, 2021). These differences matter because alternative operationalizations may lead to different identifications of involuntarily immobile or "trapped" populations and even inconsistencies when not considered properly (Ayeb-Karlsson et al., 2018) .

Beyond the challenges associated with choosing the appropriate level of and dimensionality when operationalizing migration capabilities, a further difficulty concerns the temporal and spatial nature of people's responses to environmental stress. Recent work emphasizes that

¹ Some studies have begun to question the assumption that voluntarily immobile populations are necessarily less vulnerable than involuntarily immobile ones, suggesting instead that they may underestimate environmental risks and remain exposed to future vulnerability (Zickgraf, 2018).

households' responses are often dynamic and nonlinear over time, as they may initially rely on in-situ adaptation strategies before later adopting migration if local coping capacities fail, provided they retain the capability to move (McLeman, 2018). Similarly, Rodríguez-Peña (2025) argues that the factors shaping migration aspirations and capabilities are not constant and may exert both positive and negative influences which evolve over time. In a similar vein, Schewel (2020) suggests that immobility should not be treated as a static residual category, but rather as a dynamic process unfolding across time and space, meaning that an individual considered immobile under one geographical lens or period may not necessarily be so under another. Another challenge is, as Rodríguez-Peña (2025) highlights, that migration aspirations and capabilities are not always explicitly articulated by individuals themselves, making certain forms of immobility conceptually artificial (Rodríguez-Peña, 2025). Beyond the challenges discussed by Rodríguez-Peña (2025), existing operationalizations may also face difficulties in capturing how mobility and immobility are experienced unevenly across social groups (Zickgraf, 2021)

All these observations raise important empirical and methodological questions regarding how climate-related (im)mobility should be studied. In response, this paper compares different operationalizations of mobility capabilities across varying levels of aggregation and temporal dimensions. The aim is to examine how alternative operationalizations may lead not only to different identifications of mobility-constrained or involuntarily immobile populations but also different policy approaches. To explore these issues empirically, the paper draws on data from three waves of the Mexican Family Life Survey (MxFLS) panel and is organized into four sections. Following the introduction, the second section reviews some of the main operationalizations of mobility capabilities found in the empirical literature. The third section applies selected operationalizations to the Mexican case using the available panel data. The final section presents the conclusions and preliminary policy recommendations.

2. Review methodology and research questions

To guide the retrieval of studies for this review, the main research question was defined using the Population–Concept–Context (PCC) framework, as presented in Table 1. This process yielded the following central research question:

1. How has the capability to migrate been operationalized in the climate-related (im)mobility theoretical frameworks?

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Table 1.

Population – Concept – Context (PCC) framework application

Population	Households and individuals exposed or vulnerable to climate-related shocks / Trapped populations
Concept	Operationalizations and indicators of mobility capabilities
Context	Climate-related (im)mobility and environmental migration

Source: Author

Building on this main question, the study is further guided by the following secondary questions:

2. What gaps do existing operationalizations present in capturing the spatial, temporal, and unequal nature of mobility capabilities?
3. How do selected operationalizations of capabilities apply to the Mexico data and which trade-offs does this reveal?

To identify studies relevant to these questions, a literature review of articles published between 2011 and 2026 was conducted using Scopus, ScienceDirect (50 studies), and Google Scholar, complemented through backward snowballing. Following Sengupta (2025), 2011 was selected as the starting point because the Foresight Report published that year helped establish immobility and “trapped populations” as topics within migration research. Due to time constraints, the review is limited to approximately 30–35 studies. Search strings are presented in Annex 1.

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3. Limitations of the paper

The limitations of this paper stem from its scope, time constraints, and the nature of the dataset used to explore the research questions. First, although the aspirations–capabilities framework conceptualizes immobility as the result of both migration aspirations and mobility capabilities, this paper focuses primarily on problematizing the operationalization of the latter. This is not to say that aspirations are free from conceptual ambiguities, as several authors have shown (Black & Collyer, 2014; Carling & Schewel, 2019; Schewel, 2020; Zickgraf, 2018) but addressing both dimensions simultaneously falls beyond the scope of this review. Similarly, meta-analyses on climate-related (im)mobility (see Beine & Jeusette (2018)) suggest that differences in findings are also shaped by contextual conditions, the measurement of climate stressors, econometric

methodologies, and the frequency of mobility data, aspects which also fall beyond the scope of this paper. It is also beyond the scope of this paper to empirically examine all potential operationalizations proposed in the literature. Therefore, only a selected set of measures will be explored using the MxFLS data, namely two unidimensional indicators (employment status and educational attainment) and one multidimensional operationalization.

Second, while some studies examine mobility capabilities at broader structural or national scales, the micro-level nature of the dataset used in this paper limits the empirical exploration to individual, household, and community-level dimensions. In addition, because mobility capabilities are understood as dynamic and evolving constructs, the three-wave structure of the dataset may not fully capture long-term responses, particularly to slow-onset environmental change. Existing literature also suggests that immobility should be understood relationally, as the (im)mobility decisions of some household members may shape the mobility capabilities of others or what Ayeb-Karlsson et al., (2018) terms “enabling immobility”. However, quantitatively operationalizing these interrelational dynamics remains methodologically difficult. Finally, because mobility capabilities constitute a latent construct that may only become observable when migration emerges as a concrete possibility, some underlying mechanisms may not be fully captured through quantitative data alone.

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4. Literature review

This literature review is structured in four parts. First, it outlines the evolution of the main theoretical approaches used to understand climate-related immobility, as these later shape the indicators used to operationalize mobility capabilities empirically. Second, it compares how these approaches conceptualize mobility capability across different levels of analysis (i.e. individual – households). Third, it reviews the main proxies and indicators used to identify mobility-constrained and voluntarily immobile populations. Finally, it critically examines the extent to which these operationalizations capture the spatial and temporal dynamics of climate-related (im)mobility.

Commented [GL4]: Another important limitation concerns how different studies classify the factors shaping mobility capabilities. For example, some authors conceptualize social capital as a direct component of mobility capability itself, while others treat it more as a “conversion factor” that influences how individuals are able to transform other resources into actual mobility opportunities.

4.1. Evolution and competing theoretical approaches to (im)mobility

Building on the reviews by Ayeb-Karlsson et al. (2018) and Zickgraf (2021), approaches to climate-related immobility can be broadly classified into three stages: (a) neoclassical approaches,

(b) expanded neoclassical approaches, and (c) agency-oriented approaches. According to these authors, the literature has progressively shifted from explaining immobility mainly through financial constraints toward emphasizing the role of agency and decision-making. Early studies, strongly influenced by the Foresight report, understood immobility primarily as the result of resource-related barriers such as asset loss, poverty traps, or insufficient financial capital. Later approaches expanded this perspective by incorporating legal and social constraints (i.e. people are not just trapped in places but also in situations), while also emphasizing that vulnerability to involuntary immobility is unevenly distributed across social groups. In this regard, the focus of analysis gradually shifted from identifying “what” constrains mobility to understanding “who” is disproportionately constrained. More recent agency-oriented approaches further challenge the assumption that immobility necessarily reflects an inability to migrate, highlighting that some individuals may voluntarily remain in environmentally vulnerable areas due to emotional, cultural, or place-based attachments that extend beyond purely economic forms of decision-making.

Within this broader evolution, three frameworks are particularly useful to conceptualize (im)mobility outcomes in the face of environmental degradation: i) the New Economics of Labour Migration (NELM) which aligns more closely with earlier resource constraint approaches, ii) the aspirations–capabilities framework which aligns with both assumptions of the second and third stage, and the mobilities paradigm, which aligns with the third stage as well but further expands attention toward the dynamic and relational nature of mobility (Zickgraf, 2021). As will be discussed below, these frameworks differ not only in their assumptions regarding mobility and immobility, but also in their preferred levels of analysis, ranging from household-centered to more individual or relational perspectives

4.1.1. New economics of labour migration (NELM)

Originally proposed by Stark & Bloom (1985) the New Economics of Labour Migration theory shifted the focus from individuals as the primary migration decision-making unit to households as the decision-making unit, conceptualizing migration as a collective strategy to diversify income through risk spreading, while responding to broader market failures (Massey et al., 1993). Therefore, migration was not longer seen as an individual-driven rational response to wage differentials in the labour market. In regards to its application, Zickgraf (2021) notes that, although NELM had not been directly applied to studies of trapped populations or involuntary

immobility, there was growing interest in understanding migration as part of an array of adaption strategies in response to environmental change. More recently, however, Letta et al., (2024) explicitly applied NELM together with the notion of poverty trap thresholds to empirically identify climate immobility traps in Nigeria.

Most importantly, even when not directly used to estimate immobility traps, studies applying NELM to households in environmentally stressed contexts still provide insights into how the “immobility paradox” may emerge. Particularly, NELM is able to shed light into this paradox by unpacking the interactions between mobility and immobility processes within households (Sengupta, 2025). Recent studies such as those of Fréguin-Gresh & Cortès (2026) and Loebach, (2016) in the context of Nicaragua, and that of Pakravan-charvadeh et al., (2025) for Iran illustrate how NELM helps explain bifurcated household responses to climate change, whereby some members migrate while others remain in place to minimize the occurrence of a particular risk, for example, food insecurity or livelihood collapse. At the same time, these studies also help clarify how households with different characteristics may display different preferences or capabilities for (im)mobility. Loebach (2016), for example, finds that households owning businesses were less likely to adopt migration strategies after Hurricane Mitch hit, instead relying on a “hands-on-deck” approach in which all members remained to sustain local livelihoods. Importantly, the author suggests that these households could still be considered capable of migrating, thereby illustrating a potential form of voluntary immobility.

Additionally to the points illustrated above, Zickgraf (2021) discusses a series of both positive and negative aspects to adopting NELM for explaining immobility. A first positive aspects is that NELM avoids casting household members who stay behind as completely passive actors devoid of agency because they are at least able to (in principle) exercise some form of influence by being part of a broader household well-being strategy and by potentially sharing its benefits. Second, just like with the example of business owning households in Loebach (2016), NELM can also illustrate which kinds of households are prone to becoming involuntarily immobile, recognizing that migration is not an available option for all households, specially impoverished ones who may prioritize urgent survival. Despite these benefits, a key limitation of NELM is that it tends to overlook inter and intra-household power relations and inequalities that shape who has the capability to migrate and who remains behind (Zickgraf, 2021). This omission is largely driven by NELM’s assumption that households are a harmonious collective unit where

all members agree on migration decisions, that is who should migrate and how benefits (i.e. remittances) should be distributed (Zickgraf (2021) citing De Haas & Fokkema, 2010).

4.1.2. The aspirations-capability framework

The second, and perhaps most influential, theoretical framework used to understand climate-related (im)mobility is de Haas's (2021) aspirations–capability framework, which builds upon Carling's aspirations–ability model by arguing that capability, rather than simply ability, is central to understanding migration and immobility. Drawing heavily on Amartya Sen's capability approach, it conceptualizes migration as a two-level process shaped by both aspirations and capabilities to migrate within perceived geographical opportunity structures. Within this perspective, human mobility represents people's freedom to choose where to live, meaning that both voluntary mobility and voluntary immobility are understood as expressions of the same decision-making agency.

According to de Haas (2021), the framework emerged in response to earlier migration theories that struggled to simultaneously account for agency and structural constraints, thereby failing to explain why migration often originates not from the poorest populations but from relatively better-off groups. Similarly, the framework critiques earlier functionalist approaches, including NELM, for overlooking how unequal access to mobility opportunities may reproduce existing inequalities, while also departing from historical-structural perspectives by incorporating individual agency through migration aspirations. Consequently, the framework proposes three outcomes resulting from the interaction between aspirations and capabilities: mobility (possesses both), voluntary immobility (capability with not aspirations), and involuntary immobility (aspirations with no capabilities), with a fourth category, “acquiescent immobility” added by Schewel (2020), referring to when individuals are devoid of both.

Since its proposal, a diverse number of studies on (im)mobility have used this approach to understand the complex relationships between climate change and migration. One example is that of Freihardt's (2025) study in Bangladesh, which directly applies the framework to analyze involuntary immobility. The author argues that environmental degradation may simultaneously increase migration aspirations while disproportionately eroding their capabilities, thereby generating forms of involuntary immobility (see **Figure 1**). The study also incorporates de Haas's argument that aspirations and capabilities are not fully separate dimensions but rather interact with each other. Additionally, Black & Collyer (2014) make an important contribution by

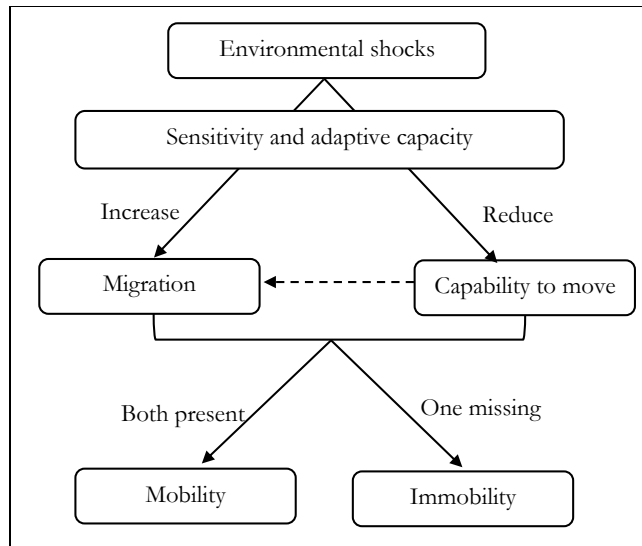
suggesting that what distinguishes trapped populations in environmental contexts from broader forms of involuntary immobility is the notion of need, as migration aspirations alone may not be sufficient to classify someone as environmentally trapped (Zickgraf, 2021)². At the same time, some authors emphasize that mobility capabilities should not be conflated with capabilities to remain in place, a distinction that has received increasing attention in recent research (Zickgraf, 2021).

Apart from offering alternative explanations for how involuntary immobility may emerge following climate shocks, the aspirations–capability framework also helps address some limitations of NELM-type approaches by placing greater emphasis on individual-level aspirations and capabilities. In doing so, it highlights how unequal power relations within households shape migration and immobility outcomes, where some household members (mainly women) remain behind to enable the mobility of others by, for example, assuming care responsibilities (Zickgraf, 2021), what is termed as “enabling immobility” by Ayeb-Karlsson et al., (2018) or as “care-based immobility” by Boldrini & Saggiomo (2026). Another example is Vlasov's (2025) study on Roma women, which argues that patriarchal household structures can limit their migration capabilities by restricting their access to resources, while also reinforcing perceptions of migration as risky and inappropriate for women, leading to internalized negative perceptions, similar to what Schewel (2020) terms “repel” factors.

² Although not the focus of this paper, this illustrates some of the measurement decisions and conceptual ambiguities associated with the other side of the equation, namely migration aspirations.

Figure 1

Effect channel of climate shocks on aspirations and capabilities according to Freihardt (2025)



Source: Freihardt (2025)

4.1.3. Mobilities paradigm

Finally, according to Zickgraf (2021), while NELM and the aspirations–capabilities framework are useful for understanding mobility capabilities at the household and individual levels respectively, the mobilities paradigm is better suited to capturing the complex dynamic and spatio-temporal nature of (im)mobility. Originally proposed by Sheller & Urry (2006) this approach focuses less on proposing specific indicators of mobility capability and more on understanding how different forms of mobility interact with each other (i.e. the movement of people with the movement of goods) and unfold unevenly across time and space through the interaction of environmental, social, and institutional factors (Boas et al., 2018). For example, Boas et al., (2018) noticed that rapid-onset climate shocks often generate visible and concentrated population movements that attract humanitarian attention, whereas slow-onset degradation tends to produce more gradual (in time), fragmented (in space), and less visible mobility patterns (non-massive), despite their long-term consequences.

Therefore, this perspective draws attention to the relative nature of (im)mobility across both time and space. In this regard, an individual responding gradually to slow-onset environmental degradation may appear relatively immobile when compared to someone displaced rapidly following a sudden disaster, without necessarily being immobile in an absolute sense. Although the mobilities paradigm does not propose specific indicators or a particular level of analysis through which to measure this relativeness, it raises important questions regarding the temporal and spatial scales at which mobility capabilities should be evaluated. Similarly, questions remain regarding the spatial scale at which mobility itself should be understood, such as whether migration capability is reflected only through long-distance or international migration, or also through shorter local displacements or even everyday movements.

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The frameworks discussed above, rather than representing competing explanations, should be understood as complementary perspectives on climate-related (im)mobility (see Table 2). As Massey et al. (1993) and Zickgraf (2021) suggest, migration research lacks a single unified theory and instead relies on a fragmented set of approaches that illuminate different dimensions of mobility and immobility, each contributing differently to the understanding and operationalization of mobility capabilities. As (Sengupta, 2025) implies, choosing one or the other should depend on the type of question the researcher seeks to answer.

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Table 2

Comparative synthesis of frameworks explaining climate-related (im)mobility

Comparison aspect / Framework	NELM	Aspirations-capability	Mobilities paradigm
Focus	Household risk adaptation strategies	Interaction between aspirations and capabilities	Dynamic and relational nature of mobilities
Level at which capability is conceptualized	The household	The individual	Not explicit
Unit of analysis	The household	The individual	Mobility processes and the interactions between them.

Understanding(s) of immobility	Enabling immobility / Failed household local (on-farm) and external (off-farm) adaptation	Low aspirations and/or lack of capability	Relative and dynamic in a spatial and temporal sense, not absolute immobility
Main limitation	Overlooks intra-household inequalities by assuming equal mobility capabilities across household members, potentially underestimating vulnerability to involuntary immobility.	The measurement of capabilities still faces important challenges, including the selection of appropriate individual-level indicators. Further differentiation is also needed between capabilities to stay and capabilities to move ³	Due to its emphasis on the relative spatio-temporal nature of mobility, even short everyday movements may be considered forms of mobility, which, as Zickgraf (2021) notes, risks making immobility almost invisible.
Temporal dimension treatment	Partial, recognizes that household-level migration may emerge over time as a last resort adaption strategy after local coping strategies are exhausted.	Not explicit. However, some authors like Sengupta (2025), by proposing a continuum approach (discussed below), have expanded upon the original framework to incorporate this layer of analysis.	Central to the framework

³ Returning to Loebach's (2016) example of households remaining to sustain their businesses, the same asset may simultaneously increase mobility capabilities by providing migration resources while also strengthening the capability and willingness to stay.

Types of capabilities	Household index	assets	Employment status	Not applicable
operationalizations	Household value	assets	Education level	

Source: Author

4.2. Unidimensional operationalizations of migration capabilities

As already suggested, the theoretical frameworks discussed above have important implications for how mobility capabilities are operationalized empirically. It's important to clarify that, due to the emergent nature of the field, many operationalizations remain author-specific proposals rather than standardized indicators like those found in studies related to food security, for example. To organize the literature, this review distinguishes between unidimensional and multidimensional operationalizations of mobility capabilities. Unidimensional approaches capture only one dimension of mobility constraints, such as financial, human, social, political, or natural capital. In contrast, multidimensional operationalizations aggregate several dimensions into a single composite indicator.

4.2.1. Financial dimension

4.2.1.1. *Employment status*

In individual-level studies of immobility, which often employ the aspirations–capabilities framework, one of the most common proxies for migration capability is employment status. For example, DeWaard et al., (2022) use this indicator to identify immobile populations in Mexico, assuming that as migration requires economic resources, having a job provides individuals with them. However, two important limitations could emerge, particularly in developing country contexts. First, employment may not be the only means through which migration is financed. For example, a recent IOM (2019) report documents that migration in many regions is frequently financed either through wage-deduction arrangements, whereby migrants repay migration costs through deductions from future earnings in destination jobs, or through loans obtained from informal lenders, relatives, or microfinance institutions. Consequently, access to migration financing may not depend exclusively on current employment status, but also on individuals' access to informal credit mechanisms and social support networks.

Second, employment status alone may inadequately capture effective mobility capability in contexts where informality and precarious labour conditions are widespread. For example, the

latest ILO world outlook report estimates that for 2024, 88.5% of people in low-income countries worked in the informal sector. For instance, if some groups of people are under or over represented in jobs with precarious conditions this proxy may under or over estimate their vulnerability to involuntary immobility. In a similar argument, Villarreal & Blanchard (2013) show that the retain potential of jobs is shaped not only by employment status itself, but also by non-pecuniary characteristics of the jobs. As an example, the authors find that some types of informal labour, such as self-employment, confers people with more autonomy, which in turn, reduces their desire to migrate. By extension, this could suggest that different forms of employment may shape migration aspirations and capabilities in distinct ways. More broadly, whether this indicator overestimates or underestimates trapped populations or incorrectly identifies them could ultimately depend on how employment conditions and access to other means of financing migration, are distributed across the sample.

4.2.1.2. *Household's asset ownership*

From studies at the household level, which often use NELM as theoretical framework, household's asset ownership has been used often as a proxy.

4.2.2. Human capital dimension

4.2.2.1. *Education level*

4.2.3. Natural capital dimension

4.2.3.1. *Soil quality*

4.3. Multidimensional operationalizations of migration capabilities

4.3.1. Individual level index with psychological and physical factors

This index author is Obi (2021)

4.3.2. Individual level index with inter-temporal focus

Drawing on Schewel (2020), Hosnedlová and Maidanik (2025) develop an index that is particularly relevant because it incorporates often overlooked dimensions of migration capability, including learned behavioural patterns and prior migration experience. Although the article was not specifically intended to explain climate immobility, it offers important insights into the determinants of migration capability.

4.3.3. Country level index with weighting scenarios

This index author is Rodríguez-Peña (2025)

Commented [GL7]: Comment: I know that this indicator is built using the PCA method, and, as such, can be considered multidimensional in an statistical sense. However, for this review, multidimensional proxies are those that consider together different dimensions of the construct, beyond just the financial aspects

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4.4. Spatial and temporal dimensions of migration capabilities

Since the emergence of NELM and the aspirations-capability framework several authors have expanded them to better incorporate dimensions related to the dynamic and relational nature of (im)mobility. Two of these contributions will be discussed below to assess the extent to which the operationalizations presented above are able to capture such complexities.

4.4.1. Coping strategies thresholds perspective

4.4.2. Continuum approach to (im)mobilities

The author of this approach is Sengupta (2025) drawing on Schewel (2020).

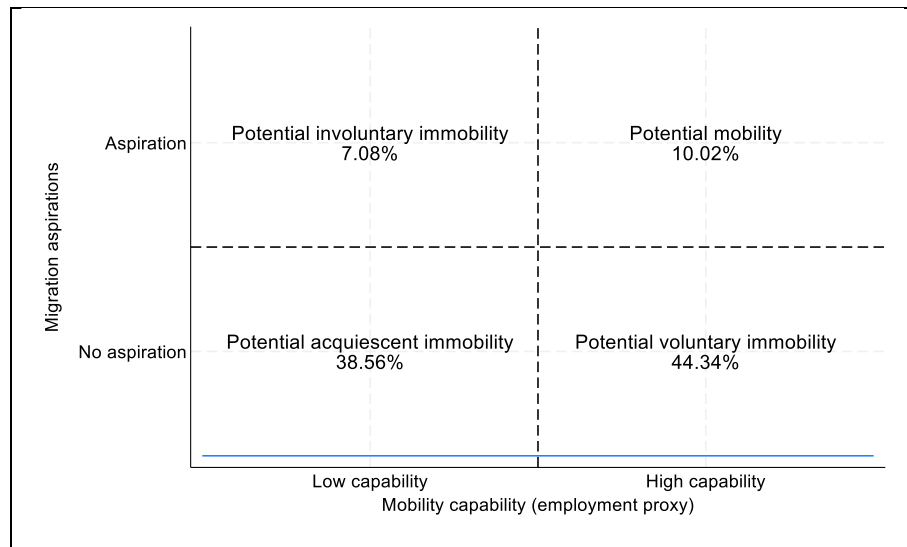
5. Illustrations with the Mexico Family Life Survey Data

In this section, I apply to the MxFLS data two unidimensional proxies of mobility capability and one multidimensional proxy, and descriptively compare the populations identified by each in order to examine the trade-offs involved in identifying involuntarily immobile populations. This analysis remains purely descriptive, and no causal interpretation should be inferred. However, it may help reveal differences across operationalizations that could influence model specification in a more detailed analysis.

Applying this first notion of capability (employment status) to the MxFLS suggests that in wave 1, 54% of the population could be considered “capable” of migrating under this unidimensional operationalization, as they reported performing a job or activity that contributed to household expenditure during the interview period. However, this result changes substantially once the quality and stability of employment are considered. A more detailed analysis reveals that 78% of employed individuals could be classified as informal workers, meaning they lacked access to social security. This pattern persists across multiple occupational categories: 65% of non-agricultural workers and 88% of rural labourers are informal. Moreover, 19.3% of employed individuals reported receiving no monetary compensation for their activity.

Figure 2

Operationalized (im)mobility categories using the employment proxy



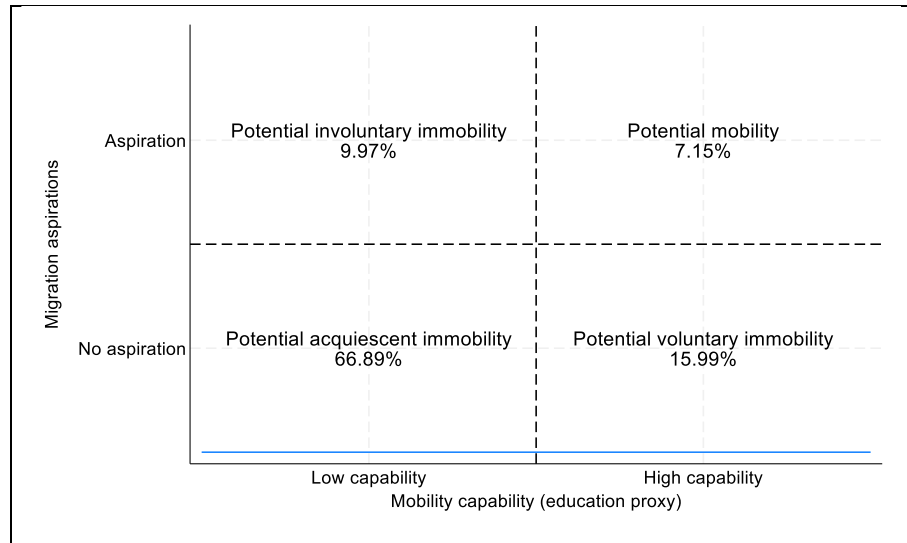
Source: Author.

Note: (Im)mobility categories are based on de Haas (2021) and Schewel (2020)

Comparing the previous unidimensional indicator with the second one, that of education, the MxFLS data reveals that only 8.56% percent of the people had college education or above, 10.20% had no instruction, and the remaining 81.76% had instruction below college level. Therefor, depending on the cutoff level chosen to define who has the capability to move according to this unidimensional operationalization , if one applies a broad cutoff where what matters is any degree of education, then 92% of people would be considered capable of moving, and if one applies a more strict measurment where only highly educated people could be considered capable of moving, that is, those with college degree, then only 8.56% would be considered capable of moving.

Figure 3

Operationalized (im)mobility categories using the education proxy



Source: Author.

Note: (Im)mobility categories are based on de Haas (2021) and Schewel (2020)

6. Conclusions and preliminary policy implications

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Annexes

Annex 1.

Search strings applied in research databases

Database	Search strings	Documents
Science Direct	("capability to migrate" OR "involuntary immobility" OR "trapped populations" OR "immobility traps")	50
Scopus	AND (operationalization OR conceptualization OR indicator) AND ("climate migration" OR "environmental migration")	149